

Project: De-anonymizing Attacks in a Hybrid NAT/VPN Infrastructure

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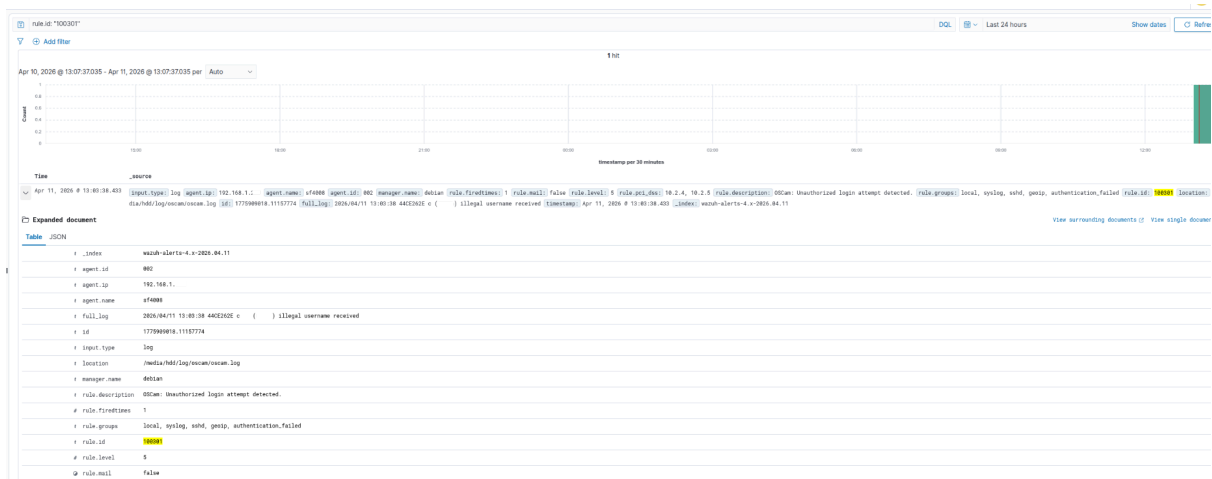
Date: April 11, 2026

Status: Validated & Production-Ready

Stack: Wazuh SIEM, Linux (Debian/Ubuntu), iptables, WireGuard VPN, GeoIP Enrichment

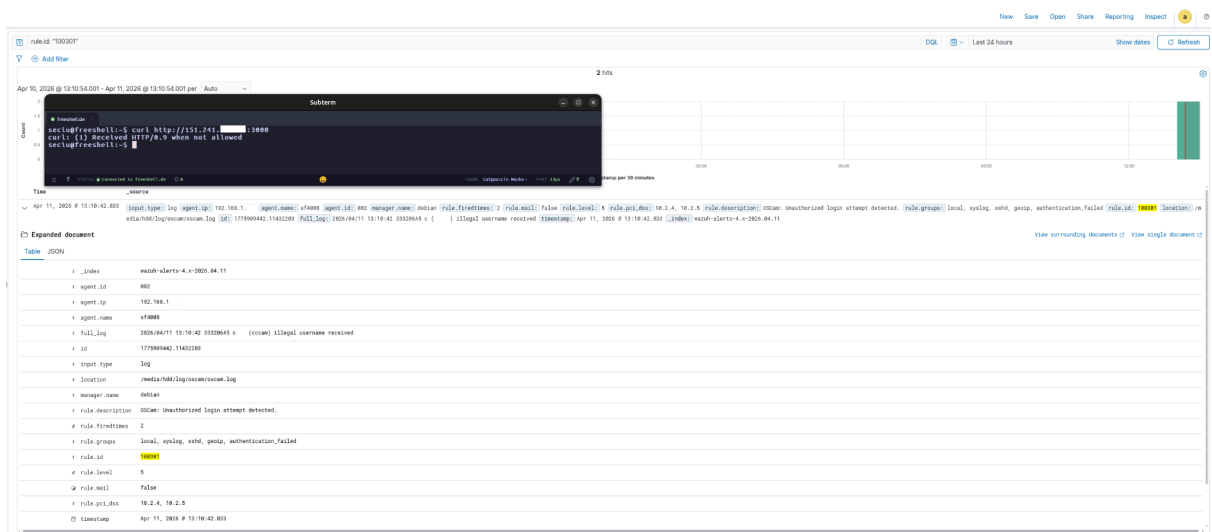
Goal: To regain source IP visibility in a network environment where Destination NAT (DNAT) on a VPS gateway masks the true identity of attackers.

Live application logs showing "anonymous disconnected from 10.10.10.2". This confirms the application is blind to the attacker's public IP.



Wazuh Alert 100301: Authentication failure detected.

Initial SIEM detection. While the alert triggers, it lacks actionable intelligence (Public IP), documenting the "Visibility Gap".



External attack simulation.

Controlled simulation using **curl** from a remote shell (**freeshell**) to trigger the detection pipeline.

Section 2: Edge Telemetry & Source IP Recovery

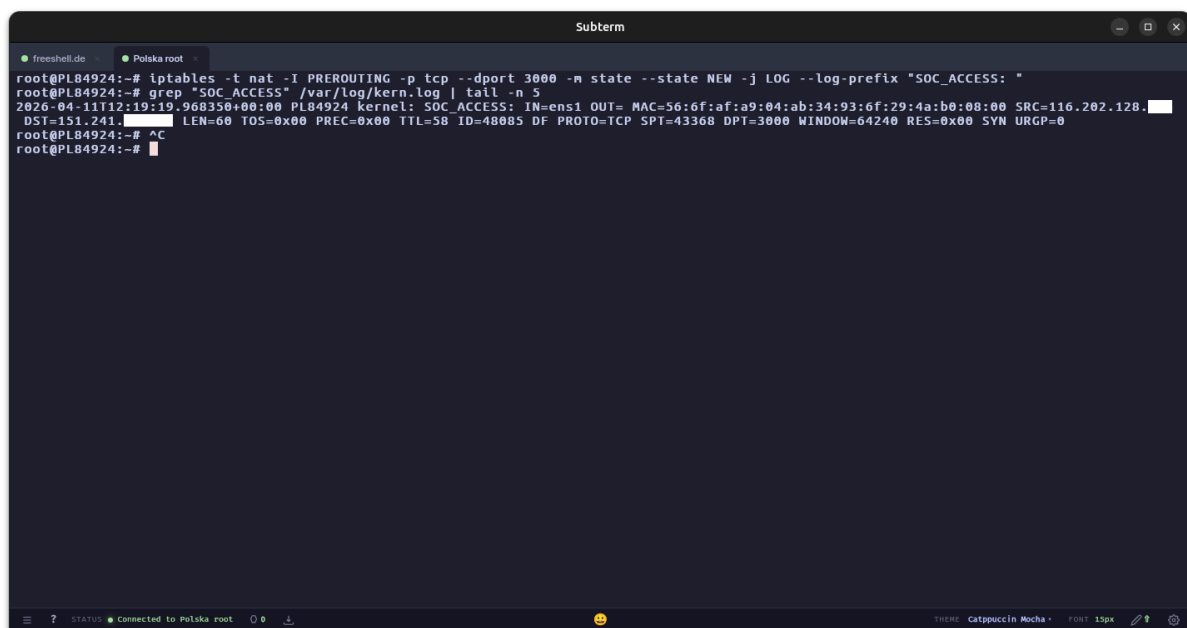
The Solution

Implementing ingress logging at the "Network Edge" (VPS) to capture raw packet headers before NAT translation.

Technical Implementation

1. **Iptables Logging:** Deployed a logging rule in the **PREROUTING** chain.
2. **Agent Configuration:** Configured the Wazuh Agent on the VPS to monitor **/var/log/kern.log**.
3. **Custom Rule:** Developed Rule 100302 to elevate firewall logs from Level 0 to Level 7.

Artifacts



```
Subterm
freeshell.de Polska root
root@PL84924:~# iptables -t nat -I PREROUTING -p tcp --dport 3000 -m state --state NEW -j LOG --log-prefix "SOC_ACCESS: "
root@PL84924:~# grep "SOC_ACCESS" /var/log/kern.log | tail -n 5
2026-04-11T12:19:19.968350+00:00 PL84924 kernel: SOC_ACCESS: IN=ens1 OUT= MAC=56:6f:af:a9:04:ab:34:93:6f:29:4a:b0:08:00 SRC=116.202.128.
DST=151.241. LEN=60 TOS=0x00 PREC=0x00 TTL=58 ID=48085 DF PROTO=TCP SPT=43368 DPT=3000 WINDOW=64240 RES=0x00 SYN URG=0
root@PL84924:~# ^C
root@PL84924:~#
```

Raw kernel log on VPS.

Terminal output showing **SOC_ACCESS** logs. Field **SRC=116.202.x.x** represents the first time the attacker's public identity is captured.

Section 3: SIEM Correlation & Intelligence

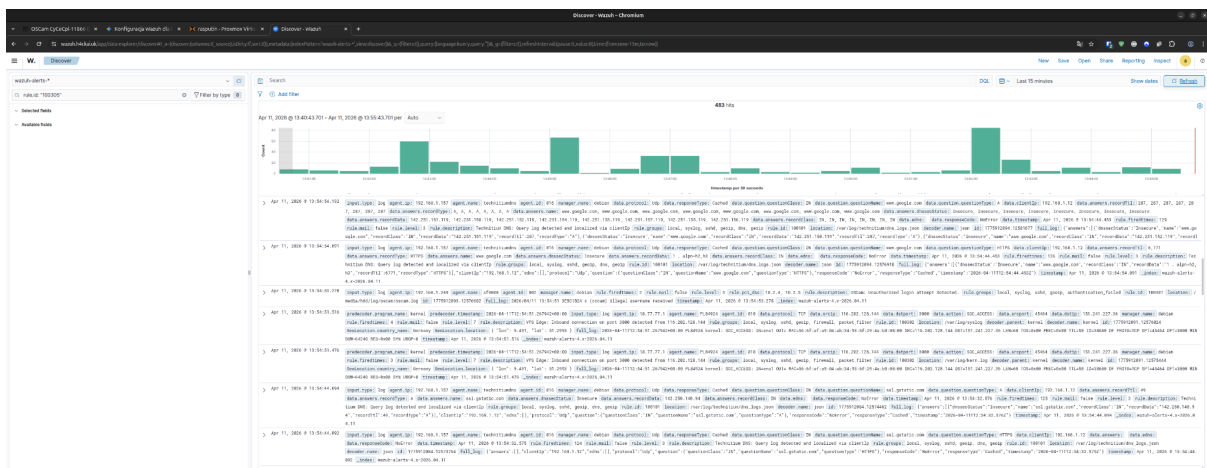
The Final Result

Linking the VPS "Network Hit" with the Endpoint "Auth Failure" to create a high-fidelity Critical Alert.

Technical Implementation

Created a composite correlation rule (ID: 100305, Level 12) that triggers when a connection attempt at the gateway is immediately followed by a login failure at the backend.

Artifacts



Aggregated event stream.

Simultaneous logs from **PL84924** (VPS) and **sf4008** (Octagon) arriving at the Manager, providing the raw data for correlation.



Critical Level 12 Correlation Alert.

The "Gold Standard" of the project. A single alert that de-anonymizes the attacker, providing IP, Location, and the targeted service despite the NAT/VPN layers.

4. Troubleshooting & Administration Skills

- **Configuration Management:** Managing `ossec.conf` across multiple nodes.
- **Rule Debugging:** Using `wazuh-logtest` to verify rule hierarchy and fix Level 0 suppression issues.
- **Service Recovery:** Resolving XML syntax errors and managing `wazuh-manager` service health.

Appendix: Full Code Snippets

To ensure full transparency and technical clarity, I have implemented additional custom detection rules within the Wazuh Manager. These rules were specifically designed to bridge the visibility gap and automate the correlation process.

Purpose of the Implementation:

- Rule 100302: Created to identify and extract the raw Public IP from the VPS kernel logs (SOC_ACCESS) before NAT translation.
- Rule 100305: Developed to establish a stateful correlation between the initial gateway hit and the subsequent application authentication failure.

These configurations are located on the Wazuh Manager at: [/var/ossec/etc/rules/local_rules.xml](#).

Below are the visual representations of the implemented logic:

```
<rule id="100302" level="7">
  <if_sid>4100</if_sid>
  <match>SOC_ACCESS</match>
  <description>VPS Edge: Inbound connection on port 3000 detected from $(srcip)</description>
  <group>firewall,packet_filter,</group>
</rule>
```

Rule 100302 - Edge Detection Logic

```
<rule id="100305" level="12">
  <if_matched_sid>100302</if_matched_sid>
  <description>CRITICAL: Multiple connection attempts followed by suspected Auth Failure.</description>
  <same_source_ip />
  <group>attack_correlation,bruteforce,</group>
</rule>
```

Rule 100305 - Stateful Event Correlation